

$$1. (1) \quad \frac{x}{16} \cdot \left(\frac{1}{1 + \frac{x^2}{16}} \right) = \frac{x}{16} \cdot \sum_{n=0}^{\infty} (-1)^n \left(\frac{x^2}{16} \right)^n = \sum_{n=0}^{\infty} (-1)^n \cdot \frac{x^{2n+1}}{16^{n+1}}$$

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \frac{1}{16} |x|^2 < 1, \quad R=4$$

$$x=4, \quad u_n = (-1)^n \frac{4^{2n+1}}{4^{2n+2}} = \frac{(-1)^n}{4}, \text{ 发散.}$$

$$x=-4, \text{ 发散 } (u_n \equiv \frac{1}{4})$$

$$\therefore \text{收敛域为 } (-4, 4)$$

$$\begin{aligned} (4) \quad \frac{1}{2} \ln \frac{1+x}{1-x} &= \frac{1}{2} (\ln(1+x) - \ln(1-x)) = \frac{1}{2} \left(\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n + \sum_{n=1}^{\infty} \frac{1}{n} x^n \right) \\ &= \frac{1}{2} \cdot \sum_{n=1}^{\infty} (1 + (-1)^{n-1}) \cdot \frac{x^n}{n} \\ &= \sum_{n=1}^{\infty} \frac{x^{2n-1}}{2n-1}, \quad x \in (-1, 1) \end{aligned}$$

$$\begin{aligned} (7) \quad \sin^3 x &= \frac{1}{4} (3 \sin x - \sin 3x) = \frac{3}{4} \cdot \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^{2n-1}}{(2n-1)!} - \frac{1}{4} \sum_{n=1}^{\infty} \frac{(3x)^{2n-1} (-1)^n}{(2n-1)!}, \quad x \in \mathcal{R} \\ &= \frac{1}{4} \sum_{n=1}^{\infty} \frac{(3 - 3^{2n-1}) \cdot x^{2n-1}}{(2n-1)!} \cdot (-1)^n, \quad x \in \mathcal{R} \end{aligned}$$

$$(10) \text{ 设 } \frac{5x-12}{(x+6)(x-1)} = \frac{A}{x+6} + \frac{B}{x-1} = \frac{(A+B)x - A + 6B}{(x+6)(x-1)} \quad \therefore \begin{cases} A+B=5 \\ 6B-A=-12 \end{cases} \Rightarrow \begin{cases} A=6 \\ B=-1 \end{cases}$$

$$\begin{aligned} \therefore \text{原式} &= \frac{6}{x+6} + \frac{1}{1-x} = \frac{1}{1+\frac{1}{6}x} + \frac{1}{1-x} = \sum_{n=0}^{\infty} (-1)^n \cdot \left(\frac{1}{6}x \right)^n + \sum_{n=0}^{\infty} x^n \\ &= \sum_{n=0}^{\infty} \left[(-1)^n \cdot \frac{1}{6^n} + 1 \right] x^n \end{aligned}$$

$R=1$, 易知 $x=\pm 1$ 时不收敛 (一般项不趋于0), $\therefore x \in (-1, 1)$

$$2. (2) f(x) = \ln(x + \sqrt{1+x^2}), \quad f'(x) = \frac{1}{\sqrt{1+x^2}} = (1+x^2)^{-\frac{1}{2}} = 1 + (-\frac{1}{2})x^2 + \frac{(-\frac{1}{2})(-\frac{3}{2})}{2!}x^4 + \dots + \frac{(-1)^n \cdot \frac{(2n-1)!!}{2^n}}{n!}x^{2n} + \dots$$

$$= \sum_{n=1}^{\infty} \frac{(-1)^n \cdot (2n-1)!!}{(2n)!!} x^{2n} + 1$$

$$f(x) = x + \sum_{n=1}^{\infty} \frac{(-1)^n \cdot (2n-1)!!}{(2n)!! (2n+1)} x^{2n+1}$$

$$(3). \int_0^x \frac{dx}{\sqrt{1-x^4}} = f(x), \quad f'(x) = \frac{1}{\sqrt{1-x^4}} = (1-x^4)^{-\frac{1}{2}} = \sum_{n=1}^{\infty} \frac{(2n-1)!!}{(2n)!!} x^{4n} + 1$$

$$f(x) = x + \sum_{n=1}^{\infty} \frac{(2n-1)!!}{(2n)!! (4n+1)} x^{4n+1}$$

11.1.

$$1. (3) \frac{1}{6\sqrt{2}} \int_{-\infty}^{+\infty} e^{-\frac{(x-a)^2}{2b^2}} dx \quad (b>0),$$

$$= \frac{1}{6\sqrt{2}} \int_{-\infty}^{+\infty} e^{-\frac{y^2}{2b^2}} dy$$

$$= \frac{1}{\sqrt{2}} \int_{-\infty}^{+\infty} e^{-\left(\frac{y}{\sqrt{2}b}\right)^2} d\frac{y}{\sqrt{2}b}$$

$$= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{2}}{2} \cdot 2 = 1$$

$\therefore$ 级数收敛且值为 1

$$(4) \int_0^{+\infty} \frac{1+x^2}{1+x^4} dx$$

$$\text{令 } u = x - \frac{1}{x}, \quad du = dx + \frac{1}{x^2} dx, \quad \therefore dx = \frac{x^2}{x^2+1} du, \quad \text{又 } u^2 = x^2 + \frac{1}{x^2} - 2$$

$$\therefore \int_0^{+\infty} \frac{1+x^2}{1+x^4} dx = \int_{-\infty}^{+\infty} \frac{x^2}{1+x^4} du = \int_{-\infty}^{+\infty} \frac{1}{x^2 + \frac{1}{x^2}} du = \int_{-\infty}^{+\infty} \frac{1}{u^2 + 2} du$$

$$= \frac{1}{\sqrt{2}} \arctan \frac{u}{\sqrt{2}} \Big|_{-\infty}^{+\infty} = \frac{\pi}{\sqrt{2}} = \frac{\sqrt{2}}{2} \pi$$

$\therefore$ 收敛且值为 $\frac{\sqrt{2}}{2} \pi$

$$(12) \int_{-1}^1 \frac{dx}{\sqrt{1-x^2}} = \arcsin x \Big|_{-1}^1 = \pi, \therefore \text{收敛且值为}\pi$$

$$(13) \int_0^{+\infty} \frac{dx}{1+x^4}, \quad \text{令 } x = \frac{1}{t}, \quad \int_0^{+\infty} \frac{dx}{1+x^4} = \int_{+\infty}^0 \frac{-\frac{1}{t^2}}{1+\frac{1}{t^4}} dt = \int_0^{+\infty} \frac{t^2}{t^4+1} dt = \int_0^{+\infty} \frac{x^2}{1+x^4} dx$$

$$\therefore \int_0^{+\infty} \frac{dx}{1+x^4} = \frac{1}{2} \int_0^{+\infty} \frac{1+x^2}{1+x^4} dx$$

由(4),

$$\text{原式} = \frac{\sqrt{2}}{4} \pi$$

$$3.(2) \int_2^{+\infty} \frac{dx}{2x+3\sqrt{x^2+1}+6} \quad \lim_{x \rightarrow \infty} \frac{\frac{1}{2x+3\sqrt{x^2+1}+6}}{\frac{1}{x}} = \frac{1}{2}$$

故发散

$$(4) \int_0^1 \frac{1}{\sqrt[3]{1-x^4}} \quad \lim_{x \rightarrow \infty} \frac{\frac{1}{\sqrt[3]{1-x^4}}}{\frac{1}{\sqrt[3]{1-x}}} = \frac{1}{\sqrt[3]{4}}$$

故收敛

$$(8) \int_0^1 \frac{\ln x}{1-x} dx \quad \lim_{x \rightarrow 1} \frac{\ln x}{1-x} = \lim_{x \rightarrow 1} \frac{\frac{1}{x}}{-1} = -1. \therefore x=1 \text{ 不是瑕点}$$

$$\lim_{x \rightarrow 0} \frac{\frac{\ln x}{1-x}}{\ln x} = 1, \quad \text{而 } \int_0^1 \ln x dx = -1, \quad (\lim_{x \rightarrow 0} x \ln x = \lim_{x \rightarrow 0} \frac{\ln x}{\frac{1}{x}} = 0)$$

故收敛

$$(10) \int_0^{+\infty} \frac{1}{x^p + x^q} \cdot (p \geq 0, q \geq 0)$$

$$\text{原式} = \int_0^1 \frac{dx}{x^p + x^q} + \int_1^{+\infty} \frac{1}{x^p + x^q} dx$$

记为 $I_1 + I_2$,

$$I_1 \text{ 收敛} \Leftrightarrow \min\{p, q\} < 1$$

$$I_2 \text{ 收敛} \Leftrightarrow \max\{p, q\} > 1$$

$\therefore$ 若 $\min\{p, q\} < 1$ 且 $\max\{p, q\} > 1$ 时收敛,
否则发散

$$4. (2) \int_1^{+\infty} \frac{\cos(3x+2)}{\sqrt{x^3+1} \cdot \sqrt[3]{x^2+1}} dx$$

$$\left| \frac{\cos(3x+2)}{\sqrt{x^3+1} \cdot \sqrt[3]{x^2+1}} \right| \leq \frac{1}{\sqrt{x^3+1} \cdot \sqrt[3]{x^2+1}}, \lim_{x \rightarrow \infty} \frac{\sqrt{x^3+1} \cdot \sqrt[3]{x^2+1}}{x^{\frac{13}{6}}} = 1$$

$\therefore$ 绝对收敛